# 6b. Course Wrap-Up

PhD Applied Methods

Duncan Webb
NovaSBE

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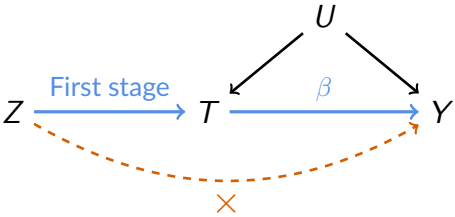

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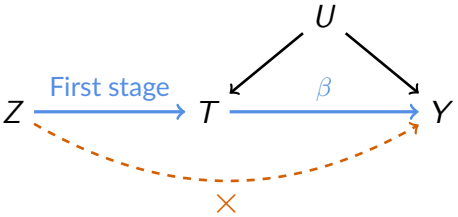
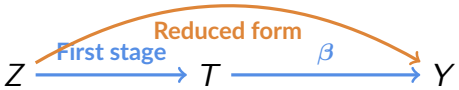



| | |
|---|---|
| $Z$ affects $Y$ **only through** $T$ | (exclusion restriction) |
| $Z$ is unrelated to $U$ | (exogeneity) |

# The Wald ratio: reduced form ÷ first stage

The DAG tells us that $Z$'s effect on $Y$ travels **entirely** through $T$:



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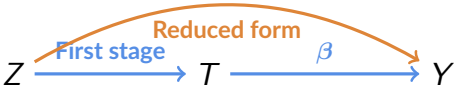


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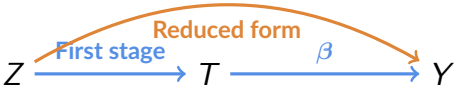


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Rearranging:

$$\beta_{\text{Wald}} = \frac{\text{Reduced form}}{\text{First stage}} = \frac{\mathbb{E}[Y_i|Z_i=1] - \mathbb{E}[Y_i|Z_i=0]}{\mathbb{E}[T_i|Z_i=1] - \mathbb{E}[T_i|Z_i=0]}$$

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| **Compliers** | 0 | 1 | Go to college *because of* scholarship |
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The instrument only **changes behavior** for compliers (and defiers). Always-takers and never-takers contribute nothing to the first stage.

# Decomposing the reduced form by group

The reduced form can be written as:

$$\underbrace{\mathbb{E}[Y|Z{=}1] - \mathbb{E}[Y|Z{=}0]}_{\text{Reduced form}} = \mathbb{E}\big[(T_i(1) - T_i(0))(Y_i(1) - Y_i(0))\big]$$

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Each group contributes according to how $Z$ changes their treatment ($\Delta T$):

$$\begin{aligned}
&= \underbrace{(+1)}_{\Delta T} \cdot \mathbb{E}[Y_i(1) - Y_i(0)|\text{Complier}] \cdot P(\text{Complier}) \\
&+ \underbrace{(0)}_{\Delta T} \cdot \mathbb{E}[Y_i(1) - Y_i(0)|\text{Always/Never}] \cdot P(\text{Always/Never}) \\
&+ \underbrace{(-1)}_{\Delta T} \cdot \mathbb{E}[Y_i(1) - Y_i(0)|\text{Defier}] \cdot P(\text{Defier})
\end{aligned}$$

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Notice: compliers get weight $+1$, defiers get weight $-1$.

This is a **non-convex** combination — the defier term enters with a negative sign!

# Monotonicity eliminates defiers

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And the first stage is simply:

$$\mathbb{E}[T|Z{=}1] - \mathbb{E}[T|Z{=}0] = P(\text{Complier})$$

# The LATE result

Dividing the reduced form by the first stage:

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**Key implications**:

- LATE $\neq$ ATE in general (unless effects are homogeneous)
- Different instruments identify different complier populations $\rightarrow$ different LATEs
- We cannot name the compliers — only estimate their share from the first stage

# Outline

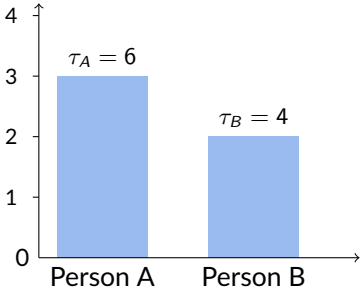
1. Recap: Instrumental Variables and the Wald Ratio

2. Why Negative Weights Break Identification

3. Course Recap

# A simple weighted average

Suppose two people each have a **positive** treatment effect:



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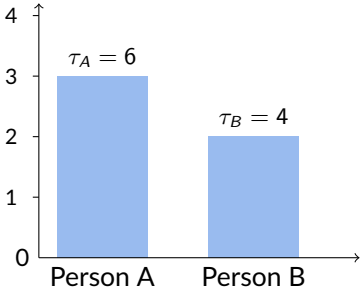


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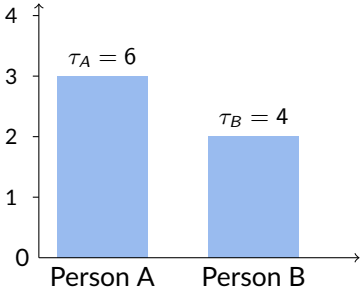


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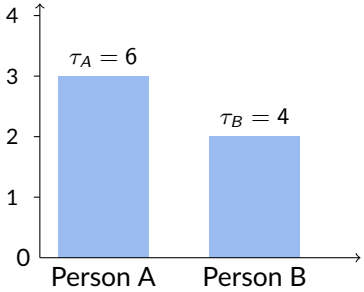


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For **any** $w_A, w_B \geq 0$ with $w_A + w_B = 1$:

$$4 \leq \bar{\tau} \leq 6$$

The average is always **between** the individual effects.

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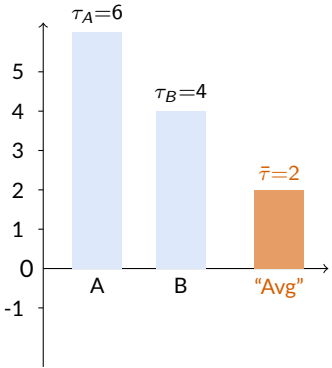

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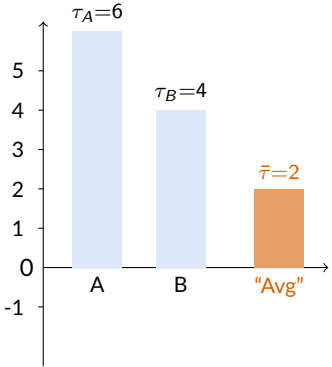


With negative weights, the "average" can be **anywhere** — even negative when all individual effects are positive!

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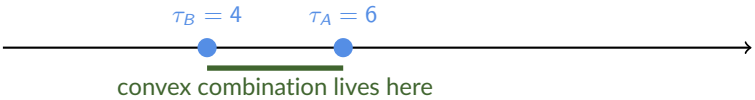


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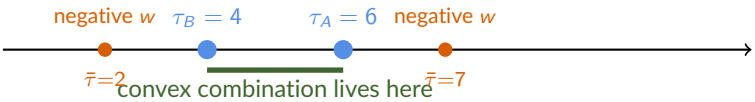


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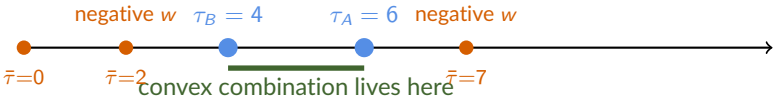

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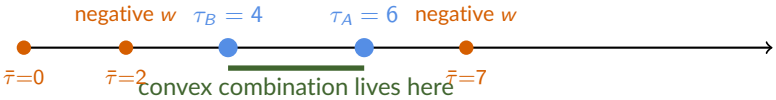


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**2. Difference-in-differences — staggered timing with TWFE**

- TWFE puts **negative weights** on some cohort-time effects when it uses already-treated units as controls
- $\hat{\beta}^{TWFE} = \sum_{g,t} w_{g,t} \cdot ATT_{g,t}$ where some $w_{g,t} < 0$
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**The fix in both cases**: ensure your estimator uses a **convex combination** of causal effects.

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2. Why Negative Weights Break Identification

3. Course Recap

# Lecture 1: Potential Outcomes and Randomization

**Core framework**: The potential outcomes model (Rubin causal model)

- Each unit has two potential outcomes: $Y_i(1)$ and $Y_i(0)$
- **Fundamental problem**: We only observe one — the counterfactual is missing
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**When compliance is imperfect**: ITT and the Wald estimator preview IV

# Lecture 2: Instrumental Variables

**Problem**: Treatment is endogenous — selection on unobservables

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**Takeaway**: IV is powerful but requires strong assumptions. Always think carefully about who the compliers are and whether the exclusion restriction is credible.

# Lecture 3: Difference-in-Differences

**Setting**: Panel data with treatment that turns on at a specific time

- Compare changes over time between treated and control groups
- **Parallel trends**: In the absence of treatment, both groups would have followed the same trajectory
- Basic $2 \times 2$ DiD; regression with TWFE (unit + time fixed effects)

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**Modern developments**:

- Event study specifications to examine dynamic effects and test pre-trends
- **Staggered timing problem**: TWFE can give negative weights with heterogeneous effects
- Solutions: Callaway & Sant'Anna, de Chaisemartin & D'Haultfoeuille, Sun & Abraham
- Synthetic DiD: combine synthetic control with DiD for better counterfactuals

# Lecture 4: Regression Discontinuity Design

**Setting**: Treatment assigned by a cutoff on a running variable $Z_i$

- **Sharp RDD**: $D_i = \mathbb{1}\{Z_i \geq z^*\}$ — compare outcomes just above vs. just below
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- Local linear regression; bandwidth selection (bias-variance trade-off)

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- Local linear regression; bandwidth selection (bias-variance trade-off)

**Fuzzy RDD**: Treatment probability jumps at cutoff but not from 0 to 1

- Ratio of jumps = LATE at the cutoff (same logic as Wald/IV)

# Lecture 4: Regression Discontinuity Design

**Setting**: Treatment assigned by a cutoff on a running variable $Z_i$

- **Sharp RDD**: $D_i = \mathbb{1}\{Z_i \geq z^*\}$ — compare outcomes just above vs. just below
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**Fuzzy RDD**: Treatment probability jumps at cutoff but not from 0 to 1

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**Validity checks**:

- McCrary density test (no manipulation of running variable)
- Covariate balance at the cutoff
- Results robust to bandwidth choice

# Lecture 5: Empirical Tools

**Practical issues** that arise across all causal inference methods:

- **Clustering**: Standard errors when observations are correlated within groups
- **Power**: Sample size calculations and minimum detectable effects
- **Attrition**: When units drop out of the sample; Lee bounds
- **Multiple testing**: Family-wise error rate corrections
- **Robustness**: Sensitivity analysis; replication
- **Spillovers**: When treatment affects control units (SUTVA violations)
- **Measurement**: Classical and non-classical measurement error

# The toolkit: matching methods to settings

| Method | Key assumption | When to use |
|---|---|---|
| **RCT** | Random assignment | You control the experiment |
| **IV / 2SLS** | Exclusion restriction | Exogenous variation in $T$ |
| **DiD** | Parallel trends | Panel data; staggered treatment |
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**Common thread**: Every method constructs a credible **counterfactual** for the treated group.

The differences are in **what you assume** to build that counterfactual, and **what population** the estimated effect applies to.

# Final thoughts

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- **Transparency**: Show your readers why your identifying assumptions are plausible — pre-trends, balance tests, placebo checks, robustness to specification
- **Know your estimand**: Are you estimating the ATE, ATT, or LATE? For which subpopulation? This shapes both the method you choose and how you interpret results